[LOGO AREA]

AMMAR ALZAIEM

CYBERSECURITY | VAPT

SECURITY ASSESSMENT REPORT

TryHackMe – Intro to LAN

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PREPARED BY	Ammar Alzaiem
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Overview

This room focuses on how devices communicate inside a local area network (LAN). It covers how devices are connected, how IP addresses are organised into subnets, and how protocols such as ARP and DHCP help devices communicate.

These are important networking fundamentals. Before working with network scanning or trying to understand traffic between devices, I need to understand how a local network is organised and how devices find each other.

1. LAN Topologies

A network topology describes how devices are connected. Different designs have different costs, advantages, and weaknesses.

The room introduces three common topologies.

Star topology

In a star topology, every device connects to a central device, usually a switch.

This design is easy to expand and troubleshoot because each device has its own connection to the centre. However, if the central switch fails, communication across the network can be disrupted.

Bus topology

In a bus topology, devices share a common communication backbone.

It is relatively inexpensive to set up because it requires less cabling and equipment. The downside is that a problem with the main backbone can affect the entire network.

Ring topology

In a ring topology, each device connects to two neighbouring devices, forming a loop. Data travels around the ring to reach its destination.

A break in the ring can disrupt communication if there is no alternative path.

The main lesson here is that network design involves trade-offs between cost, reliability, and scalability.

2. Switches and Routers

Switches and routers both help move network traffic, but they do different jobs.

A switch connects devices within a local network and uses MAC addresses to forward Ethernet frames to the appropriate ports.

A router connects different IP networks and forwards packets based on routing information.

For example, when two computers are on the same subnet, they can usually communicate directly over the local network. When a computer needs to reach a destination outside its subnet, it typically sends the traffic to its default gateway.

Understanding this difference helps when troubleshooting connectivity or mapping a network during reconnaissance.

3. Subnetting

Subnetting is the process of dividing an IP network into smaller networks. It helps organise devices and control how network addresses are allocated.

An IPv4 address contains 32 bits, commonly represented as four decimal octets.

For example:

`192.168.1.100/24`

The `/24` indicates that the first 24 bits identify the network, leaving 8 bits for host addresses.

For this example:

- **Network address:** `192.168.1.0`
- **Example host address:** `192.168.1.100`
- **Broadcast address:** `192.168.1.255`
- **Typical default gateway:** `192.168.1.1` (depending on the network configuration)

A `/24` IPv4 subnet contains 256 total addresses, of which 254 are normally available for hosts in a conventional subnet.

Why subnetting matters

Subnetting helps administrators divide networks into manageable sections. In security, separating systems into different subnets can also help limit unnecessary communication between devices.

One thing to remember is that the default gateway is not a special type of IP address by itself. It is the router or other Layer 3 device that a host uses to reach destinations outside its local network.

4. ARP — Address Resolution Protocol

ARP is used in IPv4 local networks to discover which MAC address corresponds to a particular local IP address.

Imagine a computer wants to send data to another device on the same subnet. It knows the destination IP address, but it needs the destination's MAC address to deliver an Ethernet frame.

The computer can broadcast an ARP request asking which device owns that IP address. The device with the matching address can respond with its MAC address.

The result can then be stored in the computer's ARP cache for future communication.

The distinction is important:

- IP address: Logical addressing used to deliver packets across IP networks.
- MAC address: Link-layer addressing used to deliver Ethernet frames on a local network.

ARP also has security implications. Because traditional ARP does not authenticate replies, an attacker on the same local network may attempt ARP spoofing to associate their MAC address with another device's IP address.

5. DHCP — Dynamic Host Configuration Protocol

DHCP automatically provides network configuration to clients. Without it, administrators would often need to configure IP addresses and other network settings manually.

A DHCP client typically follows a four-step process known as DORA:

1. **Discover:** The client searches for available DHCP servers.
2. **Offer:** A server offers an IP address and configuration.
3. **Request:** The client requests the offered configuration.
4. **Acknowledgment (ACK):** The server confirms the assignment.

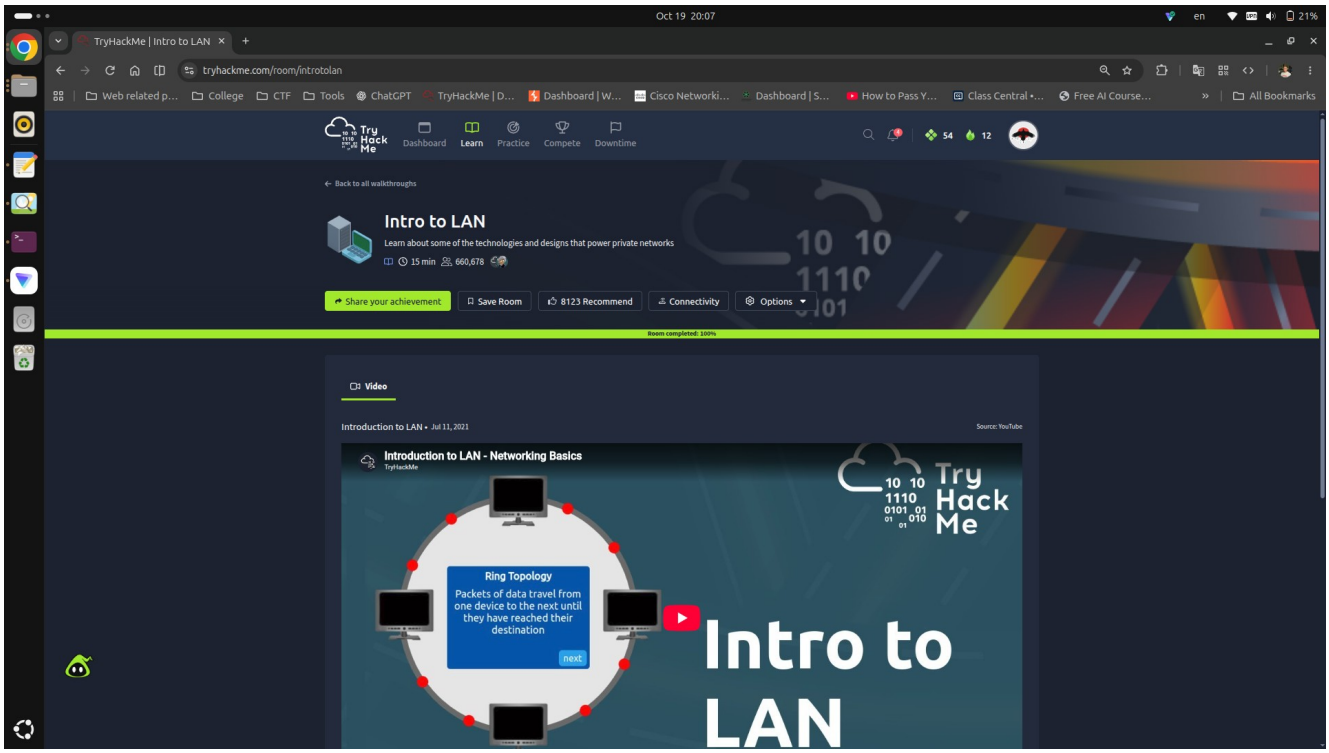
DHCP can provide information such as an IP address, subnet mask, default gateway, and DNS servers.

This makes connecting devices to a network much easier. However, DHCP also introduces security considerations: for example, a rogue DHCP server could provide clients with incorrect network settings.

Conclusion

This room helped connect the physical layout of a network with the protocols that make communication possible. Topology explains how devices are connected, subnetting defines network boundaries, ARP helps devices find each other locally, and DHCP handles network configuration.

These concepts will be useful when studying network enumeration and penetration testing. They also provide a starting point for understanding attacks such as ARP spoofing and rogue DHCP activity.


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